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Take Home Test-3

Sr. No.	Section	Task Description	CO Mapped
1	Database (MongoDB) with Backend	Design and implement Mongoose schemas in infrastructure layer for Book (title, author, ISBN, totalCopies, availableCopies), User (name, email, password, role, studentId), and Transaction (bookId ref, userId ref, issueDate, dueDate, returnDate, status, fine) supporting NoSQL flexible documents and relationships.	1
2	Database (MongoDB) with Backend	Establish MongoDB connection via Mongoose in infrastructure/config, integrate with clean architecture (injectable), handle connection errors/success with Winston logging, and use .env for DB URI and options.	1, 2
3	Database (MongoDB) with Backend	Implement Repository pattern: create domain repository interfaces (BookRepository, UserRepository, TransactionRepository), then concrete MongoDB implementations with full CRUD NoSQL operations (find, create, update, delete, aggregate) following clean architecture dependency inversion.	1, 2
4	Database (MongoDB) with Backend	Enhance repositories with transaction-specific NoSQL queries: checkBookAvailability, updateBookStockOnIssue/Return, findOverdueTransactions (using MongoDB $gt for dueDate + aggregation), and support for renew logic.	1
5	Database (MongoDB) with Backend	Add database optimizations and integrity: schema-level validations, unique indexes (email, ISBN), compound indexes for frequent queries (userId + status), and timestamps for audit trails.	1
6	Database (MongoDB) with Backend	Integrate MongoDB repositories into existing backend use-cases (replace any mocks): update application layer dependency injection, ensure all DB operations are wrapped with Winston logging for traceability and error tracking.	

POST http://localhost:3000/a

No environment

http://localhost:3000/api/auth/register

Save | Share

POST http://localhost:3000/api/auth/register Send

Docs | Params | Authorization | Headers (8) | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
{
  "name": "Bhargav",
  "email": "bhargav@gmail.com",
  "password": "password123",
  "role": "librarian",
  "studentId": "LIB001"
}
```

Body | Cookies | Headers (19) | Test Results

201 Created • 565 ms • 1 KB

{} JSON | Preview | Visualize

```
{
    "message": "User registered successfully",
    "user": {
        "name": "Bhargav",
        "email": "bhargav@gmail.com",
        "role": "librarian"
    }
}
```

POST http://localhost:3000/a | POST http://localhost:3000/a

No environment

http://localhost:3000/api/auth/login

Save | Share

POST http://localhost:3000/api/auth/login Send

Docs | Params | Authorization | Headers (8) | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

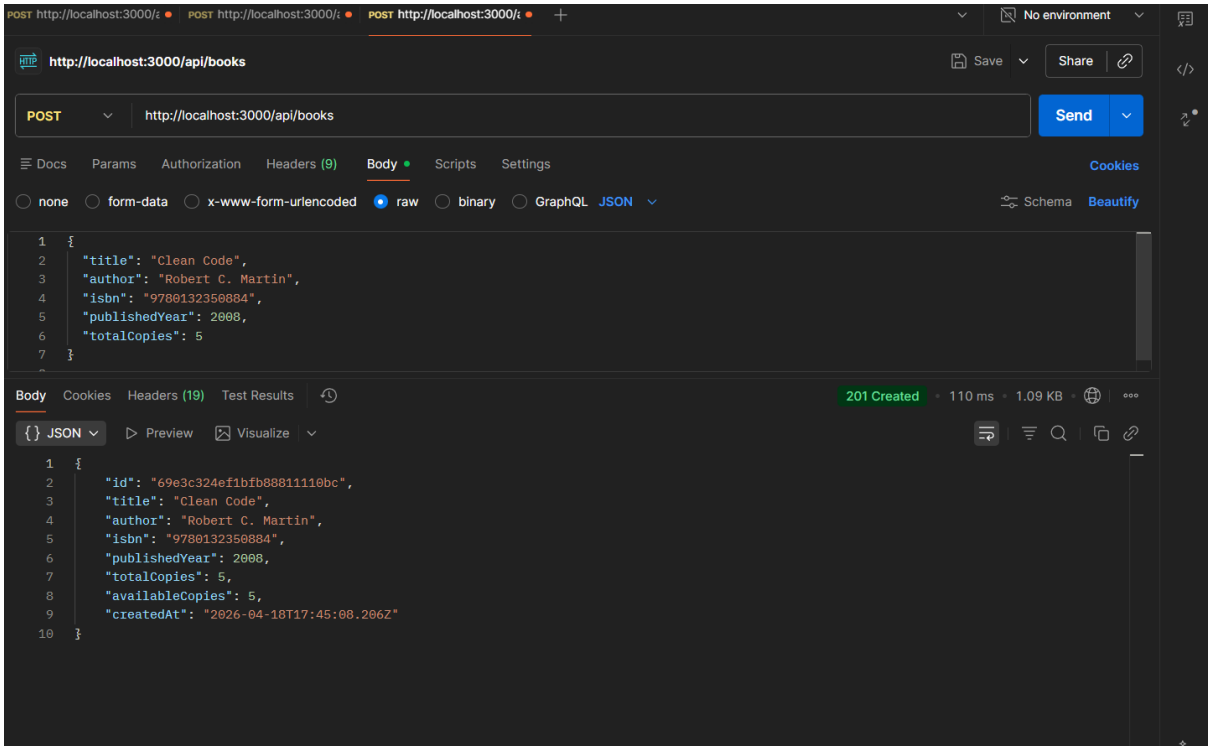
```
{
  "email": "bhargav@gmail.com",
  "password": "password123"
}
```

Body | Cookies | Headers (19) | Test Results

200 OK • 193 ms • 1.09 KB

{} JSON | Preview | Visualize


```
{
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.
        eyJpZCI6IjY5ZTNjMjdhYmE3YTZkNTY5NGJiNjA4MyIsInJvbGUiOiJsaWJyYXJpYW4iLCJpYXQiOjE3NzY1MzQyMTQsImV4cCI6MTc3OTEyNjIxNH0.
        Hn-o7TNs2wcYDUOKjewnwGuupuGJ9zzJtlrIIcH1IOU"
}
```



Log file

{"level":"info","message":"MongoDB Connected: 127.0.0.1","service":"library-management-api","timestamp":"2026-04-18 23:05:50"}

{"level":"info","message":"Server is running on port 3000","service":"library-management-api","timestamp":"2026-04-18 23:05:50"}

{"level":"info","message":"MongoDB Connected: 127.0.0.1","service":"library-management-api","timestamp":"2026-04-18 23:12:15"}

{"level":"info","message":"Server is running on port 3000","service":"library-management-api","timestamp":"2026-04-18 23:12:15"}

{"level":"info","message":"POST /api/auth/register","service":"library-management-api","timestamp":"2026-04-18 23:12:17"}

{"level":"info","message":"New user registered: bhargav@gmail.com (Role: librarian)","service":"library-management-api","timestamp":"2026-04-18 23:12:18"}

{"level":"info","message":"POST /api/auth/login","service":"library-management-api","timestamp":"2026-04-18 23:13:34"}

{"level":"info","message":"User logged in: bhargav@gmail.com","service":"library-management-api","timestamp":"2026-04-18 23:13:34"}

{"level":"info","message":"MongoDB Connected: 127.0.0.1","service":"library-management-api","timestamp":"2026-04-18 23:13:57"}

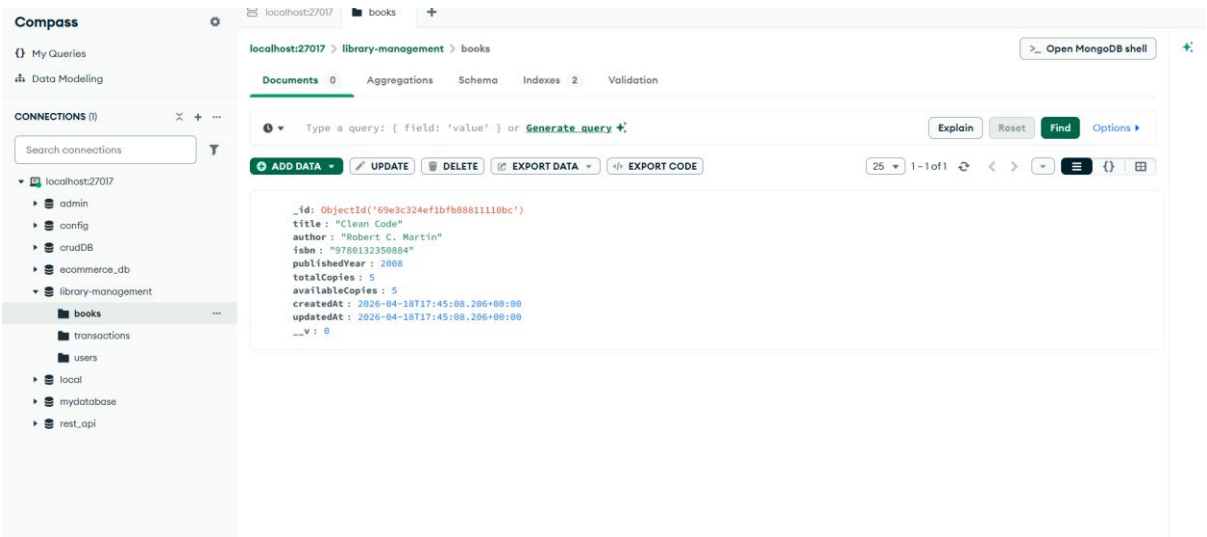
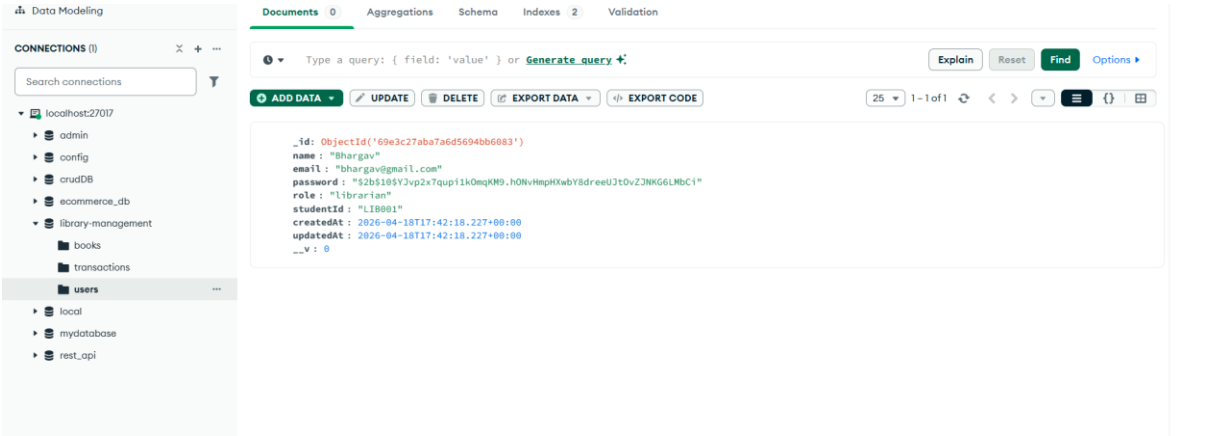
{"level":"info","message":"Server is running on port 3000","service":"library-management-api","timestamp":"2026-04-18 23:13:57"}

{"level":"info","message":"POST /api/books","service":"library-management-api","timestamp":"2026-04-18 23:14:31"}

{"level":"warn","message":"Unauthorized access attempt: No token provided","service":"library-management-api","timestamp":"2026-04-18 23:14:31"}

{"level":"info","message":"POST /api/books","service":"library-management-api","timestamp":"2026-04-18 23:15:08"}

{"level":"info","message":"Book created by Librarian bhargav@gmail.com: Clean Code","service":"library-management-api","timestamp":"2026-04-18 23:15:08"}

{"level":"info","message":"GET /api/books","service":"library-management-api","timestamp":"2026-04-18 23:15:39"}

{"level":"warn","message":"Unauthorized access attempt: No token provided","service":"library-management-api","timestamp":"2026-04-18 23:15:39"}